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Raman Scattering **Hot Paper**

Transfer Learning-Assisted SERS: Predicting Molecular Identity and Concentration in Mixtures Using Pure Compound Spectra

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Abstract: Identifying and quantifying compounds in unknown mixtures represents the ultimate goal of surface-enhanced Raman scattering (SERS) spectroscopy but remains a significant challenge in real-world applications. Existing machine learning-driven SERS methods are limited by their reliance on prior knowledge of mixture composition, while time-consuming experimental testing of all possibilities is not feasible. We integrate the molecular specificity of SERS with an adaptive transfer learning (TL) strategy to sequentially identify and quantify carnitine components in 11 unknown binary, ternary, and quaternary multicarnitine mixtures, achieving 100% identification accuracy and a mean quantitation error of only 3%. All models are trained solely on pure compound spectral data, enabling scalable, qualitative, and quantitative analysis of complex, unseen multiplex spectra—without requiring costly and time-consuming training data collection for every possible mixture. This predictive transfer learning-driven approach marks a transformative leap for practical SERS applications, allowing accurate analysis of complex mixtures without prior knowledge of components or ratios.

Introduction

Surface-enhanced Raman scattering (SERS) spectroscopy is a power tool for detecting and quantifying multiplex chemical compounds in complex real-world samples.[1–4] However, its practical application in multiplex analysis is constrained by the need for large labeled spectral datasets, which are required to establish reference libraries for identifying and quantifying individual components.[5–12] While recent advances in SERS leverage machine learning to deconvolute spectra, these methods require prior knowledge of mixture composition and large annotated datasets—an impractical requirement as mixture complexity increases.[5–12] Critically, such approaches are task-specific and fail to generalize to unknown mixtures with unidentified components and ratios, a common challenge in real-world analyses (Figure 1a).[5–12] We propose that transfer learning (TL) can address this gap by training models exclusively on pure compound spectra to extract molecular fingerprints, circumventing the need for exhaustive experimental data collection across mixture combinations. TL is a machine learning approach that repurposes pretrained models for a related task, eliminating the need to train new models from scratch. By retaining learned features through frozen initial layers, TL enables efficient adaptation to new tasks with minimal data and computational resources, a critical advantage in fields where large labeled datasets are scarce or costly to acquire.[13–17]

Herein, we integrate SERS with an adaptive sequential TL framework to identify and quantify unknown binary, ternary, and quaternary carnitine mixtures, achieving 100% identification accuracy and <3% quantitation error. Notably, our models are trained solely on pure carnitine SERS spectra, bypassing the need for mixture-specific training datasets. This framework overcomes traditional SERS limitations by eliminating dependency on extensive reference libraries and labor-intensive experimentation, offering a scalable and data-driven solution for analyzing complex mixtures with unknown components and concentrations. To demonstrate this, we train TL models on pure carnitine spectral data from structurally and spectrally similar carnitines—L-carnitine (L-car), *O*-acetyl-L-carnitine (O-car), propionyl-L-carnitine (P-car), and butyryl-L-carnitine (B-car)—each with distinct metabolic roles.[18–24] We first constructed a robust dataset by measuring the intrinsic SERS spectra of pure carnitines under applied negative electrical potentials across a wide dynamic range (10^{-5} to 10^{-9} M). Experimental spectra and in silico simulations demonstrated that applied potential enhances unique spectral fingerprints, enabling unambiguous

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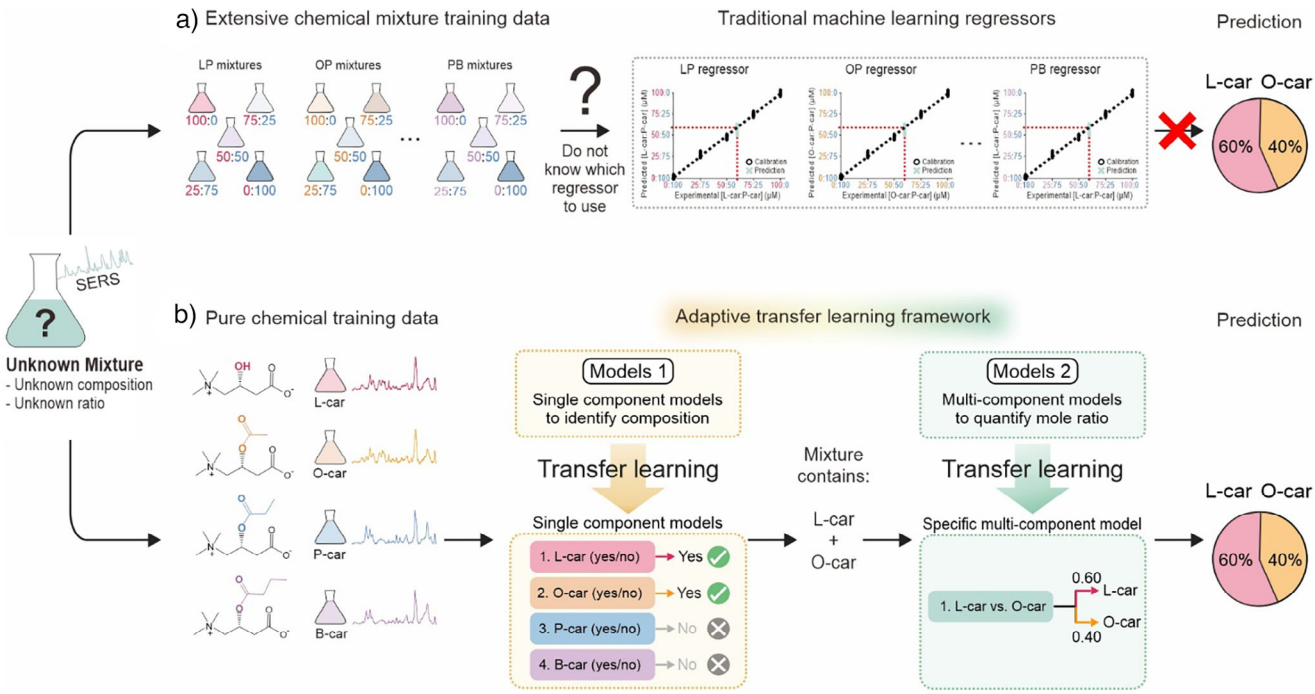

Figure 1. Schematic overview of our SERS-based transfer learning (TL) framework for unknown mixture elucidation. a) Comparison with traditional machine learning regression models, which require prior knowledge of mixture composition to build an extensive mixture dataset for model training and cannot readily be applied to predict chemical mixtures with unknown compositions and component ratios. b) Establishing an adaptive transfer learning framework to sequentially predict chemical mixtures with unknown composition and ratios. All models in the TL framework are trained on various combinations of pure carnitine spectra, retaining specific feature extraction capabilities before being applied to first identify carnitine components and then quantify their mole ratios and concentrations using untrained multiplex SERS spectra.

differentiation via principal component analysis (PCA). We leverage these pure spectra to develop an adaptive TL framework comprising two sequential model sets (Figure 1b). The first set includes four single-component random forest classifiers (RF-C) trained to distinguish individual carnitines from blank samples. These models transfer carnitine-specific spectral features to detect their presence in mixtures. The second set employs eleven RF-Cs trained on pure carnitine spectra to simulate binary, ternary, and quaternary mixtures. These models quantify component ratios using probabilities derived from spectral feature analysis. The TL framework was systematically validated against unseen multiplex spectra from 11 mixtures, achieving 100% identification accuracy and <3% quantitation error. The key innovation is an adaptive TL framework that dynamically selects optimal models for sequential identification and quantification. Single-component models first detect carnitine presence, while multicomponent models precisely quantify mixtures based on prior outputs. This stepwise and adaptive approach enhances flexibility, enabling scalability to diverse multiplex scenarios. Critically, our interpretable TL models simultaneously analyze multiple spectral features (e.g., intensity, peak position, shape), improving robustness and reducing errors compared to single-parameter methods. By employing adaptive TL, our methods transform SERS multiplex analysis from a labor-intensive process into a scalable, data-driven methodology for real-world applications, particularly where mixture components and ratios are unknown.

Results and Discussion

Spectroelectrochemical and DFT Studies of Pure Carnitines

We first measure the SERS spectra of pure L-, O-, P-, and B-carnitine analogs at their physiological concentration of 10^{-5} M. These structurally similar molecules share the same amino acid backbone but differ by their acetyl and acyl side chains (Figure 1b). Namely, L-car features a hydroxyl group attached, while O-car, P-car, and B-car possess an acetyl (CH_3CO—), a propionyl (CH_3CH_2CO—), and a butyryl ($CH_3CH_2CH_2CO$—) group, respectively. Each successive acyl modification increases the molecular size.[18,24] These carnitines are important biomarkers. For instance, elevated levels of free L-car and O-car are correlated with nonalcoholic fatty liver disease development, whereas an increase in urinary excretion of P-car is observed in methylmalonic aciduria, and underexpression of B-car is associated with primary biliary cirrhosis.[18–24] These analogous biomarkers often coexist in different ratios in biofluids and are difficult to differentiate via vibrational spectroscopy due to their similar molecular structures and spectra. Hence, it is imperative to accurately identify and quantify the individual components in complex mixtures to understand and diagnose various health conditions. To enhance SERS signals, we applied successive negative potentials (0.0 to −0.4 V) to electrostatically attract the positively charged N-terminus of zwitterionic carnitines to the negatively charged Ag nanocube-modified gold electrode

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surface (Figures 2ai and S1–S3). For example, without applied potential ($V_{applied}$), P-car exhibits only three weak bands (labels 1–3) at ~690, 1450, and 1780 cm^{-1}, corresponding to $\upsilon_{C\text{-}N+(CH3)3}$ stretching, δ_{CH} bending, and υ_{CO} stretching, respectively. In contrast, at −0.4 $V_{applied}$, seven new bands (labels 4–10) emerge at 560, 630, 800, 1000, 1100, 1300, and 1650 cm^{-1}, assigned to skeletal υ_{CC} stretching (label 4), τ_{CH2} in-plane twisting (label 5), N^+-terminus $\upsilon_{C\text{-}N+(CH3)3}$ stretching (label 6), υ_{CC} stretching (label 7), ω_{CH3} wagging (label 8), δ_{CH3} bending (label 9), and carboxylate υ_{CO2-} stretching (label 10), respectively (Figure 2ai,ii). PCA highlights the critical role of $V_{applied}$ in carnitine differentiation. Without $V_{applied}$, carnitine clusters partially overlap due to random molecular orientations and spectral similarity (Figure 2aiii). In contrast, at −0.4 $V_{applied}$, the four carnitines are well-separated with a 95% confidence, confirming the statistical significance of the spectral variances with and without $V_{applied}$ (Figure 2aiii). This separation arises from additional bands in the EC-SERS spectra, which enhances PCA variation and improves cluster resolution.

To elucidate the molecular origins of these spectral changes, we corroborate experimental spectra with DFT calculations (Figures 2b, and S4). At −0.4 $V_{applied}$, all carnitines reorient from planar to perpendicular configurations and adsorb closer to the plasmonic surface via their N^+-terminus, reducing molecular distances from 2.77–2.86 Å (no $V_{applied}$) to 2.40–2.57 Å (−0.4 $V_{applied}$) (Figures 2ai and S4).[25,26] L-car at −0.4 $V_{applied}$ exhibits six new bands (labels 4, 6–10) at 564, 744–934, 1005, 1134, 1304, and 1653 cm^{-1}, assigned to skeletal υ_{CC} stretching, N^+-terminus $\upsilon_{C\text{-}N+(CH3)3}$ stretching, υ_{CC} stretching, ω_{CH3} wagging, δ_{CH3} bending, and carboxylate υ_{CO2-} stretching, respectively (Figures 2bi and S4). O-car shows an additional band (label 5) at 634 cm^{-1} corresponds to in-plane twisting (τ_{CH2}) of its acetyl skeletal side chain alongside the six bands (labels 4, 6–10) at 561, 744–944, 1009, 1136, 1302, and 1650 cm^{-1} ascribed to the same vibrational modes as L-car above (Figures 2bii and S4). Similarly, P-car exhibits an additional band (label 5) at 635 cm^{-1} is indexed to in-plane twisting (τ_{CH2}) of its propionyl side chain, along with the same six bands (labels 4, 6–10) at 508, 743–926, 1006, 1136, 1302, and 1646 cm^{-1} (Figures 2biii and S4). B-car displays a band at 631 cm^{-1} (label 5) that also originates from in-plane twisting (τ_{CH2}) of its butyryl side chain, along with the six bands (labels 4, 6–10) at 571, 725–924, 1003, 1134, 1301, and 1646 cm^{-1} as seen in all other carnitines, well-matching the DFT calculations (Figures 2biv and S4). Our experimental, PCA, and DFT results confirm that EC-SERS mitigates spectral noise and heterogeneity by directing carnitine adsorption on the plasmonic surface, ensuring consistent carnitine-specific SERS readouts over wide concentration ranges of 10^{-5} to 10^{-9} M (Figure S5).

TL Framework Step 1: Identify Carnitine Components

Carnitines often coexist in biofluids, such as urine and blood plasma, at comparable ratios within the same order of magnitude. Thus, identifying and quantifying individual carnitines in mixtures at physiologically relevant 10^{-5} M is critical.[18,21–24]

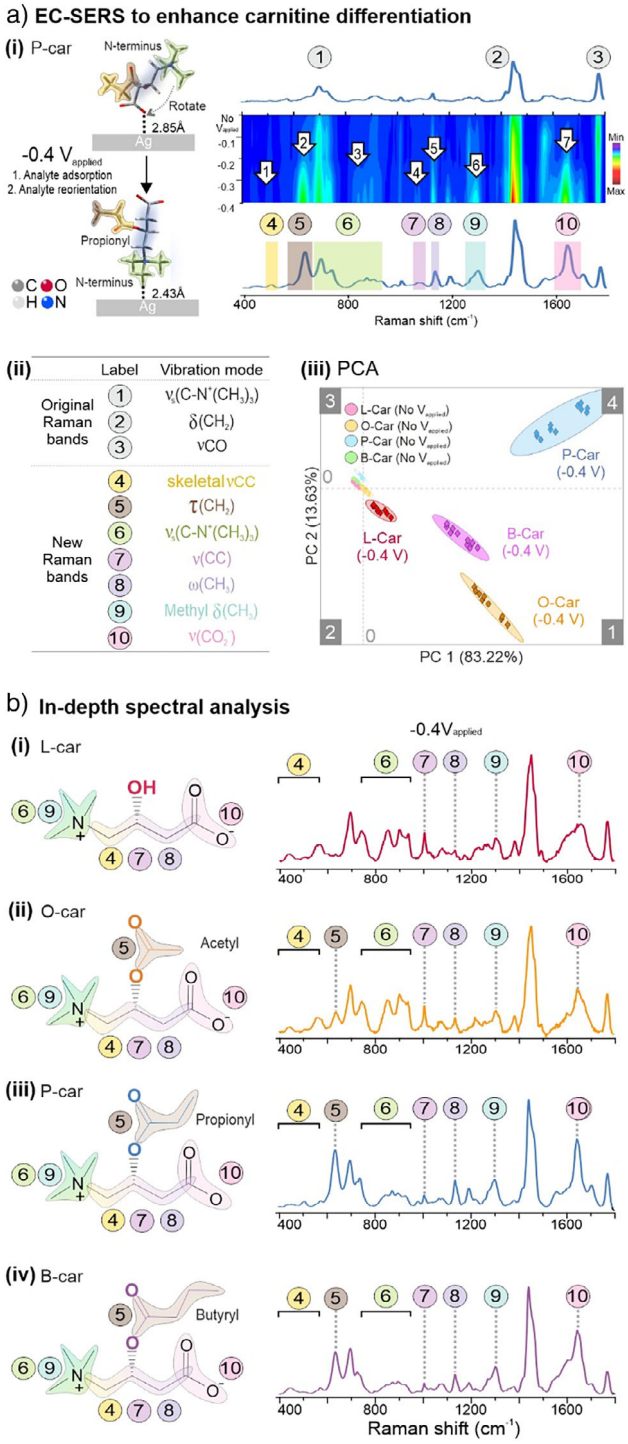

Figure 2. Electrochemical (EC)-SERS and density functional theory (DFT) study showing enhanced carnitine differentiation when a negative potential is applied. a) i) P-car adsorption and reorientation at −0.4 $V_{applied}$ and EC-SERS spectra with successive open circuit potential (no $V_{applied}$) potential and 0 to −0.4 $V_{applied}$, showing peak emergence at seven spectral regions. ii) SERS band assignment according to DFT calculations and iii) PCA showing superior differentiation of all four carnitines at −0.4 $V_{applied}$ compared to without $V_{applied}$. b) Molecular structure and experimental spectra of i) L-car, ii) O-car, iii) P-car, and iv) B-car at −0.4 $V_{applied}$ to increase spectral variations. The symbols represent the following: υ stretching, τ twisting, ω wagging, and δ bending.

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To address this, we develop an adaptive sequential TL framework trained exclusively on pure carnitine spectra. This framework integrates TL with a dynamic two-step process to accurately identify and quantify carnitines in complex mixtures (Figure 1b). In Step 1, four single-component TL classification models identify individual carnitines in an unknown mixture. Based on Step 1 results, Step 2 selects the optimal multicomponent model to quantify component ratios and concentrations. By leveraging pretrained models, our TL framework accelerates feature recognition in new datasets, eliminating the need for extensive retraining.[13–16,25] This is achieved through a process called "freezing the layers," where the model's internal layers are fixed and remain unchanged, enabling the TL models to retain critical spectral features and adapt to new tasks with minimal data and resources.[13–16,25] This allows our TL models to function as a ready-to-use fixed feature extractor detector, reusing the previously acquired learned spectral features to quickly identify important spectral features in a new task.[27,28] As a result, the model builds on what the model already learns, eliminating the need for training from scratch and saving both time and resources.

Step 1 of our TL framework identifies individual carnitine components in multicarnitine mixtures. This process involves a) extracting key spectral features from their pure carnitine spectra, b) building four single-component RF-C models to compare pure carnitine spectra to blank spectra, and c) transferring learned features to identify carnitine in multiplex spectra (Figures 3a and S6, S7). Feature importance analysis identifies the top 50 spectral features for each carnitine based on their importance (0–1 scale) and frequency across 100 train-test iterations (using color intensity) (Figures 3a and S7).[27,29,30] Selected features align with carnitine-specific SERS bands and DFT calculations, ensuring scientific relevance while filtering noise. For example, L-car identification relies on skeletal side chain's stretching (υ_{CC}, label 4), N^+ terminus stretching ($\upsilon_{C\text{-}N+(CH3)3,}$ label 6), υ_{CC} stretching (label 7), δ_{CH3} bending (label 2), and carbonyl υ_{CO} stretching (label 3) (Figure 3ai). O-car, P-car, and B-car spectral analyses emphasize features tied to their acetyl, propionyl and butyryl in-plane twisting (τ_{CH2}, label 5), along with their intrinsic skeletal side chain's stretching (υ_{CC}, label 4), N^+ terminus stretching ($\upsilon_{C\text{-}N+(CH3)3,}$ label 6), υ_{CC} stretching, (label 7), ω_{CH3} wagging (label 8), carboxylate stretching ($\upsilon_{CO2\text{-}}$, label 10), and δ_{CH3} bending (label 2) (Figure 3aii–iv). We emphasize that while O-car, P-car, and B-car share key vibrational regions, their most distinguishing features—such as wavenumber positions, intensity, and full-width half maximum—vary in 1) exact location, 2) significance for carnitine identification, and 3) frequency among the top 50 features. Subtle differences enable precise differentiation and generate unique "feature barcodes" for each carnitine, essential for identifying individual carnitines within complex multiplex spectra.

Using these selected spectral features, we trained four single-component RF-C models to distinguish individual carnitine-specific features from the blank spectra (Figures 3b and S6, S7). Trained to distinguish individual carnitine spectra from blank spectra, these models extract spectral patterns unique to each carnitine, ensuring precise identification. For example, the L-car RF-C model is trained on a binary dataset of L-car versus blank spectra, and when given an input spectrum, the model will predict whether the spectrum corresponds to L-car or blank. Once trained, the learned features are frozen to preserve their integrity for identifying L-car in mixture spectra.

After developing four single-component models, each specialized in detecting a specific carnitine, we freeze their internal layers to retain the learned spectral features. This allows precise identification of individual carnitines in an unknown mixture sample (Figure 3c). The TL models leverage spectral patterns learned from pure carnitine spectra and apply this knowledge to identify carnitine features in noisy, complex multiplex spectra. The only modification post-TL is the input data—unknown multiplex spectra replace the pure carnitine spectra used during training. Each unknown multiplex spectrum is simultaneously analyzed by the four single-component models, each of which calculates the probability of a specific carnitine's presence, with a probability > 50% indicating its detection (Figure 3c). For instance, when analyzing a mixture containing only L-car and O-car, the L-car single-component model detects L-car-related features with a 100% probability, confirming its presence. Similarly, the O-car single-component model identifies O-car-specific spectral features and predicts its presence. Meanwhile, the P-car and B-car models find no matching features, correctly predicting their absence (Figure 3c). Based on the output from Step 1 of our TL framework, we can confirm that the unknown mixture spectra contain only L-car and O-car (Figure 3c). Overall, Step 1 of our TL framework effectively bridges the gap between related tasks, enabling models trained on pure spectra to identify carnitines in mixtures accurately. By aggregating prediction results from all four models, we determine the exact carnitine composition while excluding absent components. This step is critical for selecting the appropriate multicomponent model in Step 2. Accurate identification of the mixture composition ensures precise quantification of component ratios and concentrations. Without Step 1, applying the wrong model could lead to unreliable results.

TL Framework Step 2: Quantify Carnitine Mole Ratios

Based on the exact carnitine composition determined in Step 1, Step 2 selects the appropriate multi-component model, trained on pure carnitine spectra at equal molar ratios, to quantify component ratios and concentrations in unknown mixtures. This process involves three key steps: a) identifying unique spectral features that distinguish individual carnitine under different mixture conditions, b) constructing 11 multicomponent RF-C models, and c) transferring the learned spectral knowledge to quantify molar ratios and concentrations from multiplex spectra (Figures 4 and S6, S7). Notably, Step 2 requires no additional experimental mixture spectra. Instead, models are trained exclusively on spectra from pure carnitines, simulating all possible binary, ternary, and quaternary combinations at equal molar ratios. Unlike single-component models in Step 1, which identify individual

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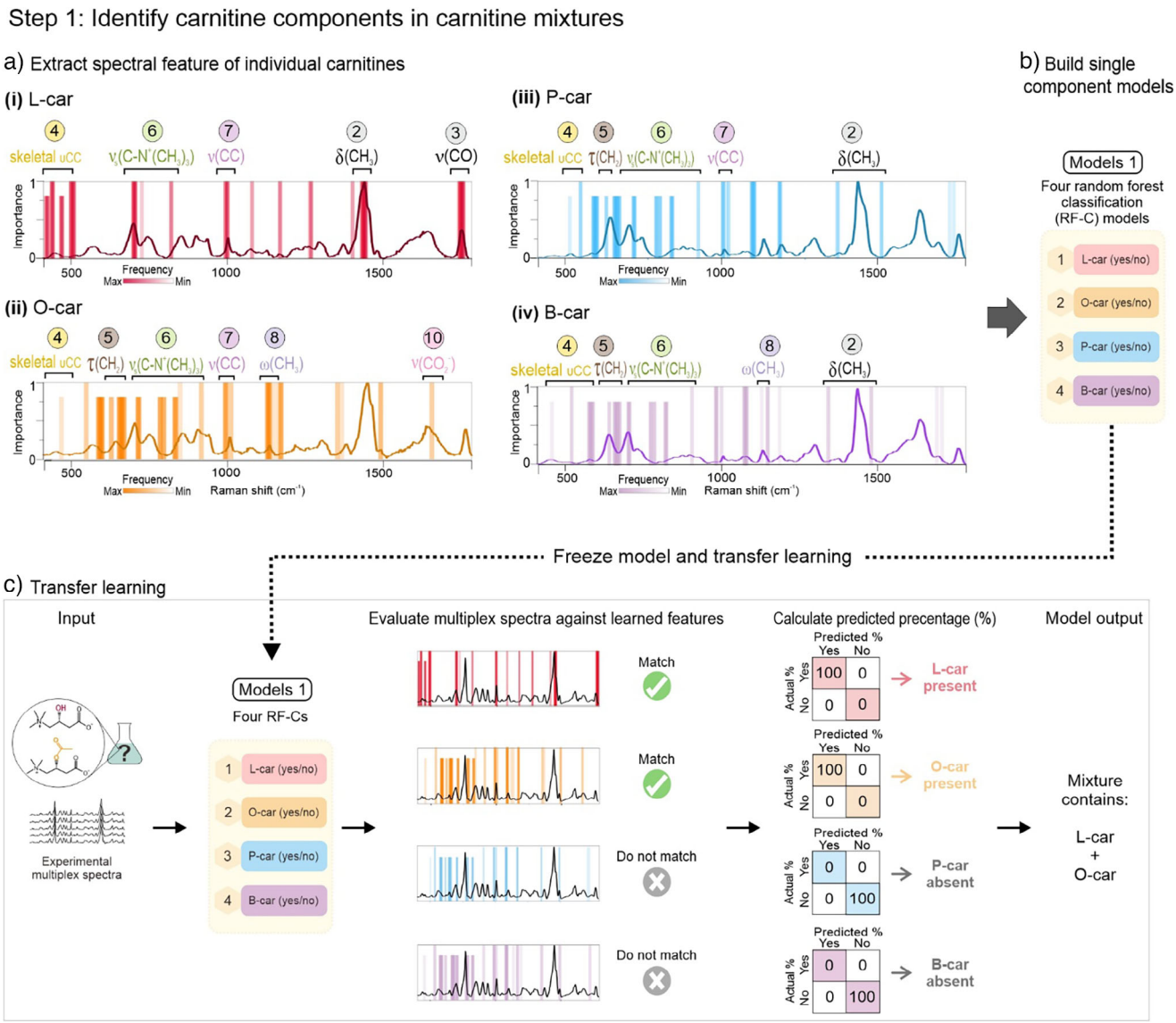

Figure 3. Step 1 of the TL framework aimed to identify carnitine components in unknown mixtures. a) Extracting carnitine-specific spectral features of i) L-car, ii) O-car, iii) P-car, and iv) B-car from their individual carnitine spectra. The feature importance barcode highlights the top 50 spectral features, scoring their importance (1 = important and 0 = not important) and selection frequency across 100 train-test iterations (dark = most selected, light = moderately selected, and colorless = not selected). Variations in height and color intensity correspond to the importance and frequency of each wavenumber feature selected during model training. b) Building four single-component models, each trained on pure carnitine and blank spectra. c) Using TL single-component models for positive carnitine identification in multiplex spectra based on spectra matching. The confusion matrix displays the accuracy of the four models when classifying five independently collected spectra of an unknown binary carnitine mixture.

carnitines by comparing their spectra to blanks, multicomponent models in Step 2 capture unique spectral differences among carnitine mixtures. These differences arise from molecular structure variation, enabling precise quantification of relative abundances of carnitine species, even when they are present in a mixture (Figures 3ai, 4b, and S7). By comparing the experimental multiplex spectra to these learned patterns, the models calculate relative abundances with high precision for accurate quantitative analysis. In contrast, single-component models cannot resolve molar ratios, as they only compare carnitine spectra to blanks individually, revealing features of each carnitine component without capturing their relative differences. This limitation can lead to missed features, signal misattribution, and inaccuracies in quantification.

We begin by performing feature importance analyses to identify key spectral features that distinguish individual carnitines based on their molecular differences under various simulated mixture conditions (Figure 3ai). For example, in a simulated binary mixture of L- and O-car at a 1:1 molar ratio, we compare their spectra across the Raman wavenumber range from 400 to 1800 cm^{-1} (Figures 4ai and S7). To evaluate the contributions of individual carnitines to the mixture spectrum, we subtract their spectra (i.e., L-car–O-car) to determine the positive contributions of each component. Subtracting O-car from L-car spectra reveals

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Figure 4. Step 2 of the TL framework to quantify carnitine components in unknown mixtures. a) Extracting key spectral differences among various carnitine combinations using their pure spectra: i) In-depth feature importance analysis of the L-car versus O-car multicomponent model for mole ratio prediction. ii) Feature importance analysis of the other 10 combinations of two to four carnitine spectra. The feature importance barcode highlights the top 50 spectral features, scoring importance (1 = important and 0 = not important) and selection frequency across 100 train-test iterations (dark = most selected, light = moderately selected, and colorless = not selected). b) Building 11 diversified multicomponent models trained on two to four carnitine spectra combinations. c) Utilizing the specific L-car versus O-car TL multicomponent model (based on carnitine components identified in step 1) to quantify component ratios and concentrations based on unseen multiplex spectra.

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that L-car contributes positively to the mixture spectrum through its intense ω_{CH3} wagging mode (label 8) (Figure 4ai). This is likely due to its more flexible molecular structure, facilitating greater vibrational freedom.[31–33] Conversely, O-car contributes positively in regions such as in-plane twisting (τ_{CH2}, label 5), N^+ terminus stretching ($\upsilon_{C\text{-}N+(CH3)3,}$ label 6), υ_{CC} stretching (label 7), and carboxylate stretching ($\upsilon_{CO2\text{-},}$ label 10). These spectral features arise from O-car's additional acetyl side chain, which introduces τ_{CH2} and intensifies $\upsilon_{C\text{-}N+(CH3)3}$ and υ_{CC} stretching due to increased polarization effects.[31–33] Collectively, this in-depth feature importance analysis highlights several spectral regions (labels 5–8 and 10) where significant spectral variations in intensity and feature exist between the two carnitines due to their intrinsic molecular differences (Figures 4ai and S8). These differences provide critical insights into their molecular variations, enabling precise quantification of molar ratios and relative concentrations. Once again, by relying on just four sets of pure carnitine spectra to predict component ratios in mixtures, we eliminate the need for extensive experimental multiplex spectra, providing flexibility and scalability for real-world applications where component compositions and ratios are often unknown.

There are 11 possible mixture scenarios among the four carnitine combinations: six binary (e.g., L-car vs. O-car), four ternary (e.g., L-car vs. O-car vs. P-car), and one quaternary (i.e., L-car vs. O-car vs. P-car vs. B-car) in our multiplex combinatorial space (Figure 4ai). We systematically analyze carnitine spectra in equimolar ratios across these combinations, performing feature analysis to build a library of spectral features that distinguish components in different mixtures. Across all multicomponent feature importance analyses, the selected features correspond to in-plane twisting (τ_{CH2}, label 5), N^+ terminus stretching ($\upsilon_{C\text{-}N+(CH3)3}$, label 6), υ_{CC} stretching (label 7), δ_{CH3} bending (label 2), and carboxylate stretching ($\upsilon_{CO2\text{-},}$ label 10), which are consistent with our experimental spectra and DFT results in Figure 2 (Figures 4ai and S8). However, these spectral features vary in their exact wavenumbers, feature importance value, and frequency among the top 50 features across the 11 mixture combinations. This differentiation generates unique feature barcodes for each carnitine combination, enhancing pattern recognition of subtle compositional variations and enabling accurate multiplex quantification in complex mixtures. By relying on four sets of pure carnitine spectra to predict component ratios in mixtures, we eliminate the need for extensive experimental multiplex spectra. This approach provides flexibility and scalability for real-world applications, where component compositions and ratios are often unknown.

Leveraging these key spectral differences and unique feature contributions, we construct 11 multicomponent RF-Cs models to simulate all possible carnitine combinations and capture mixture-specific spectral differences (Figures 4b and S6, S7). Similar to the single-component model in Step 1, the multicomponent models in Step 2 are trained on 60 spectra of each pure carnitine 10^{-5} M in equimolar ratio. However, unlike the single-component RF-Cs, which focused on carnitine identification, the multicomponent RF-Cs learn spectral patterns that differentiate carnitines for precise quantification in mixtures.

In the final step, we freeze the internal layers of these multicomponent models, which were trained on simulated equimolar conditions, and transfer them to quantify the mole ratios in multicarnitine mixtures with unknown ratios using model-calculated probabilities (Figure 4c). The molar ratio of carnitine components is determined by comparing the mixture spectral features to a pre-established set of features in the multicomponent models and then calculating their relative abundance. For example, given a multiplex spectrum confirming the presence of L-car and O-car in Step 1, the corresponding binary multicomponent model is applied in Step 2 (Figure 4c). Rather than relying on a single spectral feature, the model integrates multiple spectral signals as a "feature barcode" for robust multicorrelative analysis, ensuring accurate predictions. Take the spectral features at the 640 cm^{-1} region, for example; the intensity of the τ_{CH_2} in-plane twisting band (label 5) falls between that of pure L-car and pure O-car spectra (Figure 4c). Since τ_{CH_2} intensity is positively correlated with L-car and negatively with O-car, it is a key indicator in probability calculations (Figure 4c). The model refines its analysis using all other learned spectral features, ultimately predicting that 60% of the spectrum corresponds to L-car and 40% to O-car (Figure 4c). As a result, Step 2 of our TL framework quantifies component proportions in this mixture with 0% quantitation error, directly translating probability ratios into real-world mole ratios.[29,30] Overall, our adaptive TL framework effectively analyzes an unknown binary mixture in Steps 1 and 2, using pure compound datasets to accurately determine compositions and ratios, even when the components and their molar ratios are initially unknown.

Utilizing Transfer Learning for Unknown Mixture Elucidation

To validate the generalizability of our adaptive sequential TL framework, we systematically applied it to 11 unknown binary, ternary, and quaternary multicarnitine mixtures, all at a physiologically relevant total concentration of 10^{-5} M (Figure 5a). As a blind test, we analyze the multiplex spectra of a ternary mixture containing L-car, O-car, and B-car. In Step 1, our TL framework accurately identified the presence of L-car, O-car, and B-car while correctly excluding P-car with 100% accuracy using the four single-component models (Step 1 of the framework; Figures 5ai and S9). Identifying the exact carnitine components before quantification is critical, as it ensures the correct multicomponent model is applied in Step 2. Once the components are determined, the spectra are transferred to the corresponding ternary L-car versus O-car versus B-car multicomponent model for mole ratio determination. The model analyzes the input spectra, compares them to learned features, and assigns probability scores to predict the molar ratio as 29:22:49, corresponding to concentrations of 29, 22, and 49 μM for L-car, O-car, and B-car, respectively (Step 2 of the framework; Figures 5ai and S9). The predicted values deviate by only 1%–2% from actual concentrations, demonstrating the high accuracy of

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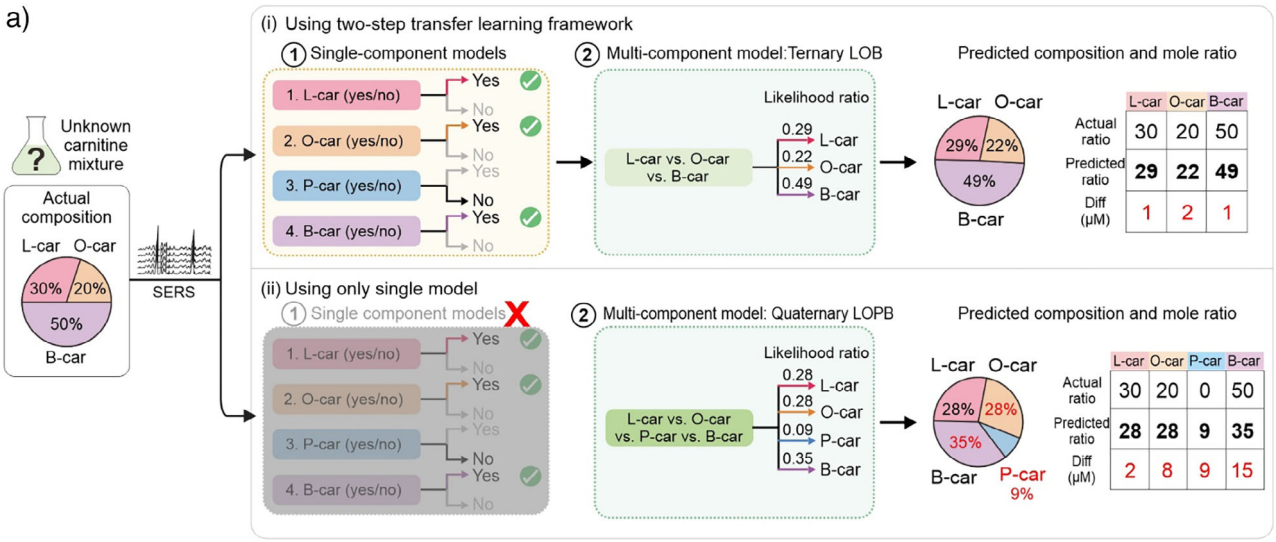

b)

		Actual		Predicted				
				Step ①		Step ②		
No.	Mixture type	Carnitines in mixture	Concentration (μM)	Carnitines in mixture	Avg Accuracy (%)	Mole ratio	Concentration (μM)	Avg difference (%)
1	Binary	L + O	50:50	L + O	**100**	46:54	46:54	**4 ± 1**
2	Binary	L + P	50:50	L + P	**100**	49:51	49:51	**1 ± 0**
3	Binary	L + B	40:60	L + B	**100**	40:60	40:60	**0 ± 0**
4	Binary	O + P	50:50	O + P	**100**	52:48	52:48	**2 ± 0**
5	Binary	O + B	60:40	O + B	**100**	56:44	56:44	**4 ± 1**
6	Binary	P + B	60:40	P + B	**100**	58:42	58:42	**2 ± 0**
7	Ternary	L + O + P	33:33:33	L + O + P	**100**	30:31:39	30:31:39	**6 ± 1**
8	Ternary	L + O + B	30:20:50	L + O + B	**100**	29:22:49	29:22:49	**1 ± 0**
9	Ternary	L + P + B	30:50:20	L + P + B	**100**	34:45:21	34:45:21	**5 ± 1**
10	Ternary	O + P + B	30:30:40	O + P + B	**100**	24:30:46	24:30:46	**6 ± 1**
11	Quaternary	L + O + P + B	25:25:25:25	L + O + P + B	**100**	25:26:30:19	25:26:30:19	**6 ± 1**
Overall					**100 ± 0 %**			**3 ± 2 %**

Figure 5. Two-step TL framework for generalized multiplex quantitation of 11 unknown binary, ternary, and quaternary carnitine mixtures. a) i) Step-by-step mixture elucidation using our adaptive TL framework demonstrates accurate carnitine identification and quantification, compared to ii) single model-predicted results for five separately prepared ternary mixtures containing L-car, O-car, and B-car as blind tests. b) Tabulated TL results show accurate and consistent elucidation of 11 binary, ternary, and quaternary carnitine mixtures in blind tests based on single-component and multicomponent models trained solely on pure carnitine spectra.

our adaptive, two-step TL strategy for analyzing unknown carnitine mixtures (Figures 5ai and S9). This precision is achieved by first narrowing down possible carnitines in Step 1 and then applying mixture-specific multicomponent models in Step 2 for quantitative analysis. We emphasize that this sequential and adaptive approach is essential to progressively handle more complex multiplex spectra, starting with single-component identification and advancing to the appropriate multicomponent model for precise quantification.

To illustrate the superiority of our TL framework, we input the same unknown spectrum into a single quaternary L-car versus O-car. versus P-car versus B-car multicomponent model—the only viable option when carnitine composition is unknown (Figure 5aii and S9). Without prior knowledge of the mixture's components, the quaternary model incorrectly detects low concentrations of P-car and struggles to accurately determine the mole ratios and concentrations of the other three carnitines, resulting in higher quantitation errors of 6%–12% (Figures 5aii and S9). These inaccuracies likely stem from spectral overlap between P-car and B-car, resulting in misclassification. Additionally, the stronger signals from B- and P-car may overshadow weaker L- and O-car bands, further complicating precise concentration determination. Overall, compared to using a single model, our adaptive TL framework demonstrates superior qualitative and quantitative prediction of ternary carnitine mixture with unknown composition and ratio.

We further validate the generalizability of our TL framework by elucidating the identity and quantity of carnitines in 11 unknown binary, ternary, and quaternary multicarnitine mixtures. These mixtures reflect the physiologically relevant concentration of 10^{-5} M, where carnitines often coexist at comparable levels in biofluids (Figure 5b). By systematically testing our dataset of 550 untrained SERS multiplex spectra (50 spectra/mixture) from 11 mixtures containing combinations of two to four carnitines, we achieve

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100% identification accuracy and <3% mean quantitation errors across all models and test instances (Figure 5b). To mitigate the risk of chance error, we randomly stratified the 50 spectral datasets (10 spectra each collected from 5 individual experiments) for each mixture using fivefold cross-validation over 100 iterations to calculate the average probability scores across all test instances. These results provide direct evidence of the generality of our adaptive TL framework, which leverages chemically relevant features to transfer knowledge from pure carnitine identification to the more complex task of multicarnitine mixture elucidation. Our approach fundamentally transforms SERS multiplex analysis by eliminating the need for multiple regression models trained on extensive multiplex spectra. Instead, it enables scalable single- and multi-component TL models trained on pure component spectra to achieve multiplex identification and quantification. This is particularly valuable when training data for new tasks is unavailable or impractical to collect, such as generating exhaustive mixture datasets covering all possible component combinations and concentration variations. This contrasts sharply with traditional multiplex regression models, which demand large volumes of multiplex data, are typically task-specific, and demand prior knowledge of mixture composition to select or train the appropriate quantification regressors.[5–12] Overall, our TL-driven SERS framework offers a powerful approach for the untargeted, ultratrace multiplex analysis of structurally similar biomarkers in unknown mixtures—an essential capability for advancing SERS toward practical sensing applications.

Conclusions

In conclusion, we have developed an adaptive TL framework trained solely on pure carnitine spectra, achieving 100% identification accuracy and a mean quantification error of just 3% across 11 binary, ternary, and quaternary mixtures. Unlike traditional regression methods, this adaptive framework efficiently identifies and quantifies components in complex mixtures without requiring extensive experimental mixture datasets. The single-component RF-C models in Step 1 accurately identify individual components, while the multi-component models in Step 2 quantify molar ratios and concentrations. Notably, our TL framework generalizes to unseen mixtures, enabling rapid, untargeted elucidation and quantification of structurally similar chemical analogs, even those with small Raman scattering cross-sections. This represents a significant advancement for practical multiplex SERS sensing. Looking ahead, we envision extending this strategy for point-of-care SERS detection in diverse applications, such as biomedical diagnostics and environmental monitoring. This could be achieved by leveraging highly sensitive platforms capable of mimicking the molecule-directing functionalities of EC-SERS. We propose creating a comprehensive SERS-based chemical space to further expand TL-driven SERS for the untargeted elucidation of other compounds beyond carnitines. This is because while the current methodology helps resolve zwitterionic carnitine species, future testing and comparative benchmarking studies across chemically diverse classes, including charged versus neutral, aromatic versus aliphatic, and polar versus nonpolar molecules, will aid in identifying systematic performance gaps and improve model generalization. This would include pure compound spectra from a wide range of small-molecule metabolites, enabling the framework to handle more complex and diverse mixtures. High-throughput automated testing platforms could facilitate mass SERS testing and flexible pre-training of TL models, allowing rapid expansion of the mixture combinatorial space for a broadened range of analytes. Furthermore, to address analytes with overlapping spectral features, we propose augmenting our framework with spectral unmixing techniques and incorporating domain-adapted feature extraction layers that can better disentangle vibrational modes appearing in the same regions.[6,8,27,28] Additionally, the impact of varying Raman scattering cross-sections can be incorporated into the framework by assigning weights based on this physical property, which would enhance prediction accuracy as the model scales to accommodate a broader range of compound classes. Finally, this TL concept can extend beyond direct SERS sensing to indirect SERS sensing, utilizing distinctive SERS probe signals to detect target analytes with weak or no Raman scattering cross-sections. The insights presented in this work represent a significant advancement for label-free qualitative and quantitative multiplex SERS sensing. By eliminating the need for extensive labeled datasets and enabling the analysis of unknown mixtures, our TL framework has the potential to transform applications ranging from nano- to bio-analytical chemistry and biomedical diagnosis. This approach concurrently addresses current limitations in SERS multiplex analysis and paves the way for scalable, data-driven methodologies in real-world sensing applications.

Supporting Information

The authors have cited additional references within the Supporting Information.[34–38]

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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